than those with healthy controls. Both of these metrics could be precisely estimated using commercially-available accelerometers.

In 2014, a Taiwanese team led by Kun-Chan Lan showcased how their pedestrian dead reckoning-based gait monitoring system could be used for the early diagnosis of Parkinson's disease.

This was followed by Kostikis et al's work on building a practical smartphone-based tool to accurately assess upper limb tremor in those suffering from Parkinson's disease.

Associated with this tremor analysis trend has been a slew of studies that have looked at replacing expensive EMG readings with smartphone accelerometric readings.

Two years ago, a team of researchers at University of Coimbra showed that measurements from off-the-shelf apps, such as iSeismometer, could be used as a reliable alternative to EMG for tremor frequency assessment when it comes to the care of patients with a diagnosis of Parkinson's, essential tremor and Holmes' tremor.

This was followed by another study published last year, where accelerometric readings from an iPhone inserted into a patient's socks could help detect orthostatic tremor. This is indeed a strong vindication of this approach, as orthostatic tremor was hitherto considered to be one of the few tremor conditions that required an EMG lab assessment for definitive diagnosis, given that leg tremors might not be visible to the naked eye.

Conclusion

On the concluding note, it is particularly impressive to note that, unlike similarly promising nexuses between gene editing and AI, for example, there is a far lesser degree of uncertainty and speculation here with noteworthy advancements already achieved that now await careful productisation.

The other two aspects that only enhance the prospects of this impending revolution of sorts are:

Emergence of deep learning as the computational framework of choice. Many of the results above have been achieved using simple (shallow) machine learning and traditional model-based signal processing approaches. As seen in the domains of speech, vision and text, deep learning has shattered state-of-the-art metrics by a fair distance. Given that adoption of deep learning-based approaches has been slow in the sensor domain, we will surely see the accuracies of these tests only go higher.

Sampling rates of sensors on phone. As things stand, motion sensor chips that are currently available in smartphones are severely underutilised in terms of the sampling rates these can provide. This is done mainly to extend battery life as well as on account of a few security concerns.

However, with the right security policies in place and use of specialised and well-monitored SDKs (such as Samsung health SDK), one can unlock the hidden potential of these nifty little workhorses, to open avenues for more granular and sensitive measurements at much higher sampling rates. This will, in turn, propel the way to more medical applications. EFY